

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO4-AT1-Test Case Design Assignment

**Case Study Title: Smart Waste Collection System
(SDG 11: Sustainable Cities and Communities)**

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Date of Submission: 18-08-26

1. Introduction

The Smart Waste Collection System is a software solution designed to improve municipal waste collection. The system monitors the fill level of waste bins, identifies bins that require collection, generates optimized collection routes, and records collection activities. The system helps reduce overflowing bins, unnecessary collection trips, fuel consumption, and operational costs.

The purpose of testing is to verify the correctness, reliability, security, usability, and efficiency of the proposed system under normal, invalid, boundary, and exceptional conditions.

2. Functional Requirements

ID	Functional Requirement
FR01	The system shall allow authorized users to log in.
FR02	The administrator shall be able to register waste bins.
FR03	The system shall monitor the fill level of waste bins.
FR04	The system shall identify bins that require collection.
FR05	The system shall generate alerts for critically full bins.
FR06	The system shall generate optimized collection routes.
FR07	The system shall allow workers to update collection status.
FR08	The system shall record completed collection activities.
FR09	The administrator shall be able to view collection history.

3. Non-Functional Requirements

ID	Requirement
NFR01	The system should provide fast response to user requests.
NFR02	The system should allow only authorized users to access protected functions.
NFR03	The system should operate reliably without frequent failures.
NFR04	The system should be easy for administrators and collection workers to use.
NFR05	The system should display accurate bin and collection information.

4. Test Scenarios

Scenario ID	Test Scenario
TS01	Verify login using valid credentials.

TS02	Verify login using invalid credentials.
TS03	Verify registration of a new waste bin.
TS04	Verify monitoring of bin fill level.
TS05	Verify identification of bins requiring collection.
TS06	Verify critical-level alert generation.
TS07	Verify invalid fill-level input handling.
TS08	Verify collection route generation.
TS09	Verify collection status update.
TS10	Verify completion of waste collection.
TS11	Verify collection history.
TS12	Verify unauthorized access prevention.

5. Detailed Test Cases

Assumption:

- Fill level **0–79%** → Collection not required
- Fill level **80–94%** → Collection required
- Fill level **95–100%** → Critical collection/alert

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Valid Login	Enter username and password → Click Login	Valid username/password	User is successfully logged in	Not Executed	Not Executed
TC02	Invalid Login	Enter incorrect password → Click Login	Valid username + wrong password	Login is rejected and error message is displayed	Not Executed	Not Executed
TC03	Register Waste Bin	Enter bin ID and location → Click Save	BIN101, valid location	New bin is registered successfully	Not Executed	Not Executed
TC04	Normal Fill Level	Enter/update bin fill level	50%	Fill level is accepted and	Not Executed	Not Executed

				collection is not required		
TC05	Collection Threshold	Update bin fill level	80%	Bin is identified as requiring collection	Not Executed	Not Executed
TC06	Critical Fill Level	Update bin fill level	95%	Critical alert is generated	Not Executed	Not Executed
TC07	Invalid Fill Level	Enter an invalid fill level	110%	System rejects the value and displays validation message	Not Executed	Not Executed
TC08	Generate Collection Route	Select bins requiring collection → Generate Route	BIN101, BIN102, BIN103	Suitable route is generated for selected bins	Not Executed	Not Executed
TC09	Update Collection Status	Worker opens assigned route → Updates status	In Progress	Collection status changes to In Progress	Not Executed	Not Executed
TC10	Complete Collection	Worker selects completed bin → Clicks Complete	BIN101	Collection is recorded as completed	Not Executed	Not Executed
TC11	View Collection History	Open collection history → Select date/bin	BIN101	Correct previous collection information is displayed	Not Executed	Not Executed
TC12	Unauthorized Access	Normal user attempts to access admin function	Normal user account	Access is denied	Not Executed	Not Executed

6. Test Design Techniques

6.1 Equivalence Partitioning

Equivalence Partitioning divides input values into valid and invalid groups.

For the bin fill level:

Partition	Range	Example	Expected Result
Invalid	Less than 0%	-10%	Reject
Valid	0–79%	50%	No collection
Valid	80–94%	85%	Collection required
Valid	95–100%	98%	Critical alert
Invalid	Greater than 100%	110%	Reject

This ensures that representative values from each input category are tested.

7. Boundary Value Analysis

Boundary Value Analysis checks values at and around important limits.

Test Value	Expected Result
-1%	Reject
0%	Accept
79%	Collection not required
80%	Collection required
81%	Collection required
94%	Collection required
95%	Critical alert
100%	Critical alert
101%	Reject

This helps identify errors occurring at input boundaries.

8. Decision Table Testing

The system decides the required action according to the bin fill level.

Condition / Action	Rule 1	Rule 2	Rule 3
Fill level	<80%	80–94%	95–100%
Collection Required	No	Yes	Yes
Critical Alert	No	No	Yes

Priority	Normal	Normal	High
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Example:

- 50% → No collection
- 85% → Collection required
- 97% → High-priority collection + critical alert

9. State Transition Testing

The waste bin passes through different states during its lifecycle.

Normal → Collection Required → Critical → Collection In Progress → Collected → Normal

Current State	Event	Next State
Normal	Fill level reaches 80%	Collection Required
Collection Required	Fill level reaches 95%	Critical
Collection Required / Critical	Worker starts collection	Collection In Progress
Collection In Progress	Collection completed	Collected
Collected	Bin is emptied	Normal

This verifies that the system correctly changes the bin state according to different events.

10. Requirement Traceability Matrix

Requirement	Test Case IDs
FR01 – User Login	TC01, TC02
FR02 – Bin Registration	TC03
FR03 – Fill-Level Monitoring	TC04, TC05, TC06, TC07
FR04 – Collection Identification	TC05
FR05 – Critical Alert	TC06
FR06 – Route Generation	TC08
FR07 – Collection Status	TC09
FR08 – Collection Record	TC10
FR09 – Collection History	TC11
NFR02 – Security	TC12

This provides clear traceability between requirements and test cases.

11. Test Coverage

Requirement Coverage

Total functional requirements = 9

Requirements covered = 9

Requirement Coverage = $(9 / 9) \times 100 = 100\%$

Scenario Coverage

Total test scenarios = 12

Scenarios covered = 12

Scenario Coverage = $(12 / 12) \times 100 = 100\%$

Condition Coverage

Test Condition	Coverage
Normal Condition	Covered
Invalid Condition	Covered
Boundary Condition	Covered
Exceptional Condition	Covered
Security Condition	Covered
State Transition	Covered
Business Rules	Covered

Therefore, the proposed test suite provides comprehensive coverage of the identified requirements and major system conditions.

12. Conclusion

The Smart Waste Collection System has been analysed and a structured test case design has been prepared to verify its functionality, reliability, security, and usability. The test cases cover normal, invalid, boundary, and exceptional conditions.

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing have been applied to improve the effectiveness of testing. The requirement traceability matrix establishes a relationship between requirements and test cases, while the coverage analysis demonstrates that all identified functional requirements are addressed.

The proposed testing approach helps ensure that the Smart Waste Collection System can reliably monitor waste bins, identify collection requirements, optimize collection activities, and improve municipal waste management efficiency.